

Unified Lifelong Action Learning: A Real-Video Pipeline for Continual Learning, Distillation, and Few-Shot Meta-Adaptation on UCF101

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Abstract—We present *Unified Lifelong Action Learning* (ULAL), an end-to-end pipeline that studies how a video action recogniser can keep learning new actions over time without forgetting the old ones, trained directly on real video frames (no cached features). The pipeline (i) builds an honest, group-disjoint split that removes an 80% leakage in UCF101’s shipped split; (ii) selects a ResNet50+LSTM backbone from a five-way bake-off (93.9% on the base classes); (iii) extends it to new classes with five continual-learning methods, where rehearsal reaches 89.8% after the full stream while plain fine-tuning collapses to 26.0%; (iv) compresses the best model into a MobileNetV3-Small student that keeps 82.7% at $\sim 14\times$ fewer parameters; and (v) adapts the student to brand-new, few-shot classes with first-order MAML, where *MAML+replay* reaches the best balance (harmonic mean 0.60) and beats a matched fine-tune baseline, isolating the meta-learning effect. For every stage we explain *why* the methods rank as they do from their abilities. A final capstone notebook (ULAL) chains the best choice of each stage and is described here, pending its run.

Index Terms—continual learning, catastrophic forgetting, action recognition, knowledge distillation, meta-learning, few-shot learning.

I. INTRODUCTION

Deep video models forget. Fine-tuning a trained recogniser on new action classes collapses its accuracy on the old ones—*catastrophic forgetting* [1]. Lifelong (continual) learning aims to add classes over time while keeping what was already learned. Video makes this harder than images: clips are large and temporal, so storing and replaying past data is expensive.

This paper describes one reproducible pipeline on UCF101 [8], trained *directly on real video frames*, so every number reflects the deployed model. Each stage is one notebook and answers one question: *how do we measure honestly? which backbone? how do we stop forgetting? how do we shrink the model? and how do we learn brand-new classes from only a handful of clips?* Our contributions:

- **An honest split.** We show 80% of UCF101 groups leak across the shipped splits and replace them with a group-disjoint split.

- **A controlled study** of five continual-learning methods, four distillation objectives, and a nine-arm meta-learning comparison whose fine-tune control isolates the meta-learning effect.
- **Explanations, not just tables.** For every stage we first say *why* one method should beat another from its ability, then show the measured result.

II. RELATED WORK

Continual learning. EWC [1] penalises changes to weights important for old tasks; LwF [2] keeps old-class outputs stable using the model’s own past predictions; rehearsal replays a few stored exemplars, and iCaRL [3] combines replay with distillation. We evaluate all of these on video. **Knowledge distillation.** A compact student can match a larger teacher’s softened logits [4] or its features; we compare logit and feature (cosine, MSE) objectives. **Meta/few-shot learning.** MAML [5] learns an initialisation that adapts quickly from few examples; because base classes outnumber new ones, we report the harmonic mean of base and new accuracy, the standard few-shot class-incremental metric.

III. METHOD, MODEL, AND SETUP

The recogniser applies a CNN backbone to each of a clip’s 16 frames, runs an LSTM over those frame features, and classifies from the final LSTM state. The classifier *grows* as new classes arrive (new rows appended, old rows kept), and only the last backbone block is fine-tuned. Clips are 16 frames at 224×224 , ImageNet normalised, on a single Colab T4 GPU. The group-disjoint split uses 18/3/4 groups per class. Within each stage, every compared method shares the settings in Table I, so any difference in the result comes only from the mechanism under test—that is what makes each comparison fair. Seeds are fixed and reset before every training run.

How to read the numbers. *Test accuracy* is the fraction correct on the unseen (group-disjoint) test set. *Old/New accuracy* split the score into classes learned earlier versus the newest task. For the few-shot stage, where 20 old classes

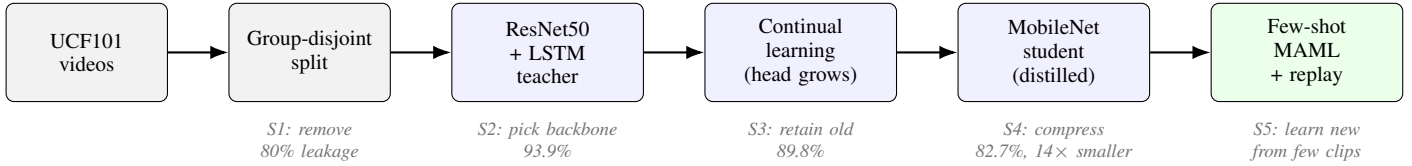


Fig. 1. The ULAL pipeline. UCF101 video is split by source group so no recording leaks across train/test (Stage 1). A ResNet50+LSTM teacher is trained on the base classes (Stage 2), extended to new classes by continual learning with a growing classifier head (Stage 3), compressed into a small MobileNet student by knowledge distillation (Stage 4), and finally adapted to brand-new classes from only a few clips with MAML (Stage 5). Each caption shows that stage’s headline result.

TABLE I
MAIN HYPERPARAMETERS PER STAGE (SHARED ACROSS THE METHODS COMPARED IN IT).

Stage	Key settings
Backbone (2)	Adam, lr 10^{-4} , 5 epochs
Continual (3)	Adam, lr 10^{-4} , 5 epochs/task; replay 15/class; LwF weight 3.0, $T=5$; EWC $\lambda=2800$
Distillation (4)	Adam, lr 10^{-4} , 15 epochs; $T=2$, CE/KD 0.75/0.25
Meta-train (5)	5-way 3-shot, 60 episodes $\times$ 8 epochs, inner SGD 5 steps @ 0.05, meta-lr 10^{-4}
Adaptation (5)	8 clips/class, 5 epochs, lr 3×10^{-4} ; replay 8/class; LwF 0.5, $T=2$

outnumber 5 new ones, a plain average is dominated by the old classes and rewards a method that simply guards them; we therefore add the *balanced* mean $(acc_{new} + acc_{base})/2$ and the *harmonic* mean, which weight new and old equally and, for the harmonic, punish sacrificing either side.

IV. STAGE 1 — DATA SPLIT AND LEAKAGE

Why this matters first. A UCF101 file `v_Class_gXX_cYY.avi` encodes a *group* gXX (one source video) and a *clip* cYY (one segment of it). All clips of a group are near-duplicate frames. If the split is made per clip, segments of the same recording land in both train and test, and the model is scored on footage it trained on—it can recognise the *scene* instead of the *action* and look far better than it is. We must therefore fix the split before trusting any later number, by assigning each whole group to a single split.

Result. Across the 20 studied classes (2,679 videos, 500 groups), the shipped split leaks **400 of 500 groups (80%)** across train/val/test; the group-disjoint split leaks **0** (Fig. 2). The disjoint split keeps the standard $\approx 72/12/16\%$ train/val/test proportions.

V. STAGE 2 — BACKBONE SELECTION

Why we expect differences. The backbone supplies the per-frame features; the LSTM only has to combine them over time. A network with no prior knowledge (a from-scratch CNN) must learn good spatial features from a few thousand clips and will struggle, whereas ImageNet-pretrained backbones already carry rich features and should cluster high. Among the pretrained ones the choice is a trade-off of accuracy, feature size, and speed, so we compare five and pick the best balance to be the teacher.

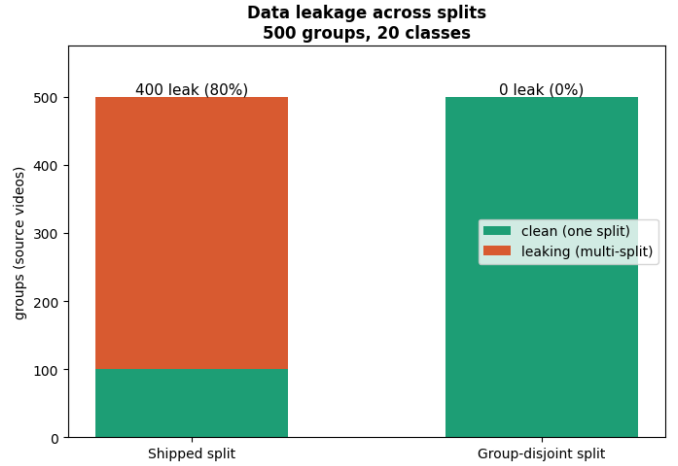


Fig. 2. Groups (source videos) that leak across splits: 80% in the shipped split versus 0% in the group-disjoint split. Every later number is read against the clean split, so the accuracies are lower but honest.

Result. Table II gives the measured numbers. The from-scratch CNN is far behind (52.8%); all pretrained backbones exceed 90%. **ResNet50 is both the most accurate (93.9%) and a good size/speed balance**, so it is chosen as the teacher for every later stage.

TABLE II
BACKBONE BAKE-OFF ON THE 10 BASE CLASSES (GROUP-DISJOINT TEST).

Backbone	Test acc. (%)	Params (M)	ms/clip
ScratchCNN	52.8	103.1	10.2
ResNet18	90.1	12.0	15.7
DenseNet121	92.0	8.3	53.1
VGG19-BN	92.0	20.8	115.7
ResNet50 (teacher)	93.9	25.9	54.6

Why the ordering. The from-scratch CNN has no prior features *and* its LSTM ingests a huge flattened feature map, giving it the most parameters (103M) yet the worst accuracy—capacity without useful priors does not help on little data. The pretrained backbones all reach the low-90s because the hard part (spatial features) is already solved. ResNet50 edges the others thanks to its deeper, richer 2048-d feature, which also gives the later distillation and meta stages a clean feature space to build on; its speed (55 ms/clip) is acceptable and far better than VGG19-BN (116 ms).

VI. STAGE 3 — CONTINUAL LEARNING

The five methods and why they should differ. Starting from the base teacher we grow the head and train on a $10 \rightarrow 15 \rightarrow 20$ class stream with each method. Their *abilities* predict the outcome: *Naive* (plain fine-tuning) has no way to protect the past, so it should forget badly; *EWC* [1] can only slow weight change with a penalty (no stored data), so it should protect some but not learn new classes freely; *LwF* [2] keeps old *outputs* stable through the model’s own past predictions (no stored data), a stronger signal than a weight penalty but still drifting; *Replay* re-exposes the model to a few real old clips, which should directly refresh the old classes; and *Replay+LwF* [3] adds output-stability on top of re-exposure. All arms share the schedule in Table I; only the mechanism differs.

Result. Table III reports accuracy after the *full* 20-class stream, split into old classes (0–14) and the newest task (15–19), plus the base (0–9) retention. Figure 3 contrasts naive forgetting with replay.

TABLE III
CONTINUAL LEARNING AFTER THE FULL $10 \rightarrow 15 \rightarrow 20$ STREAM (%).
FINAL: ALL 20 CLASSES; OLD: CLASSES 0–14; NEW: LAST TASK 15–19;
BASE: ORIGINAL 10.

Method	Final	Old (0–14)	New (15–19)	Base (0–9)
Naive	26.0	0.9	94.9	0.5
LwF	30.5	5.3	100.0	0.0
EWC	62.0	71.1	36.8	76.9
Replay+LwF	87.0	82.6	99.2	83.0
Replay	89.8	86.7	98.3	89.2

Why the winners win. The table follows exactly from the abilities. *Naive* and *LwF* learn the newest task almost perfectly (94.9% and 100%) but erase everything older (old $\leq 5\%$): with the shared weights free to move and no old data, the network re-purposes itself entirely for the last task. *LwF* even learns the new task better than naive (its distillation stabilises training) yet still forgets, because keeping *outputs* stable cannot substitute for *seeing* old data. *EWC* is the opposite failure: its penalty protects the old classes (base 76.9%) but is so restrictive that the new task barely learns (36.8%)—a stability/plasticity lock. *Replay* and *Replay+LwF* win decisively (89.8% and 87.0%) because re-exposing even 15 clips per class refreshes the old decision boundaries *while* the new task is learned. Notably plain *Replay* slightly beats *Replay+LwF* here: the buffer already protects the old classes, so the extra *LwF* distillation adds a mild drag on new-class learning without a retention gain. The lesson: *on video, storing a little data beats any data-free regulariser*.

VII. STAGE 4 — KNOWLEDGE DISTILLATION

Why and how. The best continual model is a 25.9M-parameter ResNet50 teacher; for deployment we want something far smaller. We distil it into a MobileNetV3-Small+LSTM student (~ 1.8 M parameters, $\sim 14\times$ smaller) and compare four objectives: cross-entropy only (CE), CE+KD on

softened logits [4], and CE with a cosine or a regressor-MSE loss between the 256-d LSTM states of teacher and student. We expect the student to recover most of the teacher’s accuracy (its knowledge transfers) but the gap between objectives to be small, because the student backbone is itself pretrained and already strong.

Result. Table IV: the student keeps **82.7%** of the teacher’s 87.0% at $14\times$ fewer parameters, with cosine feature-distillation the best of the four objectives.

TABLE IV
DISTILLATION INTO THE MOBILENETV3-SMALL STUDENT (20-CLASS TEST).

Model	Test acc. (%)	Params (M)
Teacher (Replay+LwF, ResNet50)	87.0	25.9
Student, CE only	80.6	1.8
Student, CE + KD	79.7	1.8
Student, CE + cosine	82.7	1.8
Student, CE + MSE (regressor)	75.9	1.8

Why. Compression works because the teacher’s decision function transfers: a $14\times$ smaller student reaches within ~ 4 points of the teacher. Among objectives, *cosine* on the LSTM state wins because it aligns the *direction* of the student’s temporal feature with the teacher’s without forcing its exact magnitude, which is a gentle, achievable target across two different backbones. Plain *KD* on logits helps less here—the student is pretrained, so labels already get it far and the extra “dark knowledge” is small. *MSE* is harshest (it demands matching magnitude too) and does worst. The honest headline is *compression* (teacher-level behaviour at $14\times$ fewer parameters), with a modest bonus from feature-cosine distillation.

VIII. STAGE 5 — FEW-SHOT META-ADAPTATION

The question and the design. Given the deployed 20-class student, how well can it learn *brand-new* classes from only a handful of clips? We keep the 20 base classes and add 5 new sports (a 5-way task, matching the 5-way meta-training). Every method uses the *same* few-shot adaptation—same 8 support clips per class, same 5 epochs, same learning rate, same seed, same replay buffer/LwF teacher—so only two things vary: the *initialisation* and whether replay/LwF are on. Three inits (*plain* student; *fine-tune*, i.e. the student after ordinary training on 30 auxiliary sports; and *MAML*, the student after meta-training on the same 30 sports) are crossed with three protection settings, giving nine arms. The *fine-tune* arm is the key control: it saw the same extra data as *MAML* but trained in the ordinary way, so *MAML vs fine-tune* isolates meta-learning from “just more data.”

Why the metric. With 20 base and 5 new classes, a plain overall accuracy is 20/25 base, so it rewards a method that guards the base and learns nothing new. We rank by the harmonic mean of base and new accuracy,

$$\text{harmonic} = \frac{2 \text{acc}_{\text{new}} \text{acc}_{\text{base}}}{\text{acc}_{\text{new}} + \text{acc}_{\text{base}}}, \quad (1)$$

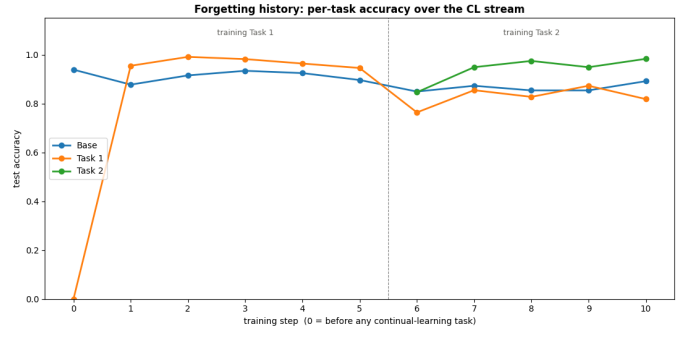
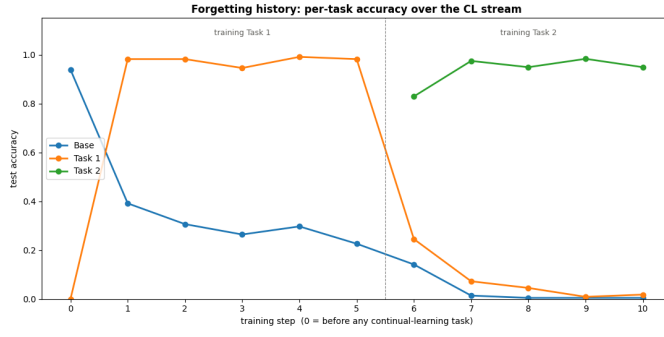


Fig. 3. Per-task accuracy across the whole stream. *Left (Naive)*: each old task’s line collapses the moment a new task is learned—catastrophic forgetting. *Right (Replay)*: rehearsing a few old clips keeps every line high.

which weights the two equally and collapses to zero if either is zero.

Result. Table V (measured; base before adapting = 0.759) ranks the nine arms. **MAML+replay is best**, reaching a balanced **0.60**, and it *beats fine-tune+replay by 0.07 harmonic*—so the gain is really from meta-learning, not merely the extra data. Figures 4–7 show the comparison, the shot-count trend, the meta-training curve, and a readable old-vs-new confusion.

TABLE V

FEW-SHOT META-ADAPTATION: 20 BASE + 5 NEW (5-WAY), IDENTICAL ADAPTATION FOR ALL ARMS, RANKED BY HARMONIC MEAN. MEASURED AT 8 CLIPS/CLASS, 5 EPOCHS.

Method	New	Base	Bal.	Harm.
MAML+replay	0.625	0.576	0.601	0.600
MAML+replay+LwF	0.580	0.542	0.561	0.561
fine-tune+replay	0.464	0.617	0.541	0.530
fine-tune+replay+LwF	0.446	0.626	0.536	0.521
replay	0.295	0.677	0.486	0.411
replay+LwF	0.223	0.724	0.474	0.341
no-meta (plain)	0.527	0.189	0.358	0.278
MAML	0.661	0.011	0.336	0.022
fine-tune	0.500	0.011	0.256	0.022

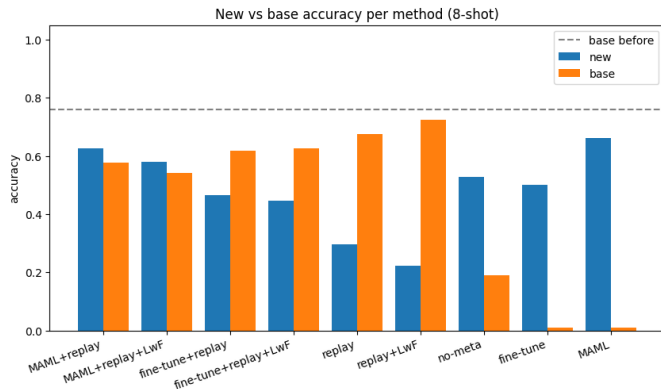


Fig. 4. New-class accuracy and base retention per method. Inits with no protection (MAML, fine-tune, no-meta) learn new classes but wipe the base; adding replay recovers the base, and the MAML init gives the highest new-class accuracy among the protected arms—so MAML+replay tops the balanced ranking.

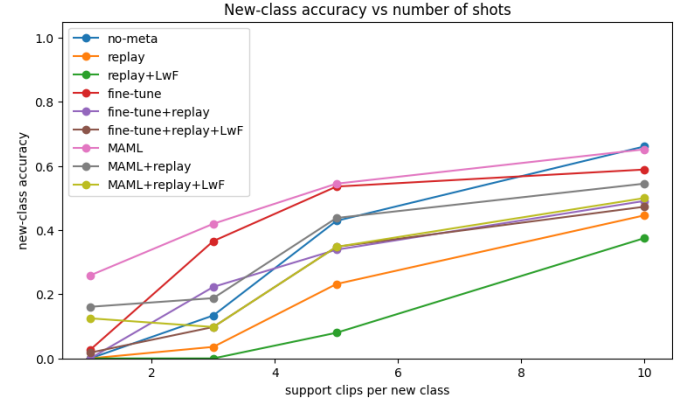


Fig. 5. New-class accuracy versus number of support clips. MAML’s advantage is largest at **1 clip** (MAML 0.26 while every other init is near 0) and narrows as clips grow—meta-learning helps most exactly when data is scarce.

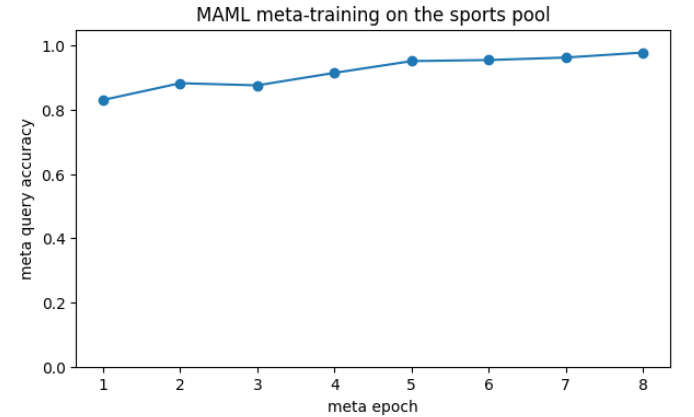


Fig. 6. MAML meta-training curve: meta query accuracy rises from 0.83 to 0.98 over 8 epochs, confirming the initialisation is genuinely being shaped, not left at the student’s starting point.

Why the winners win. Each row follows from ability. *MAML alone* has the highest new-class accuracy (0.661) but base = 0.011: a meta-learned init is built to move its features quickly onto the new task, and with no protection it moves them all the way, erasing the old classes. *Replay alone* is the

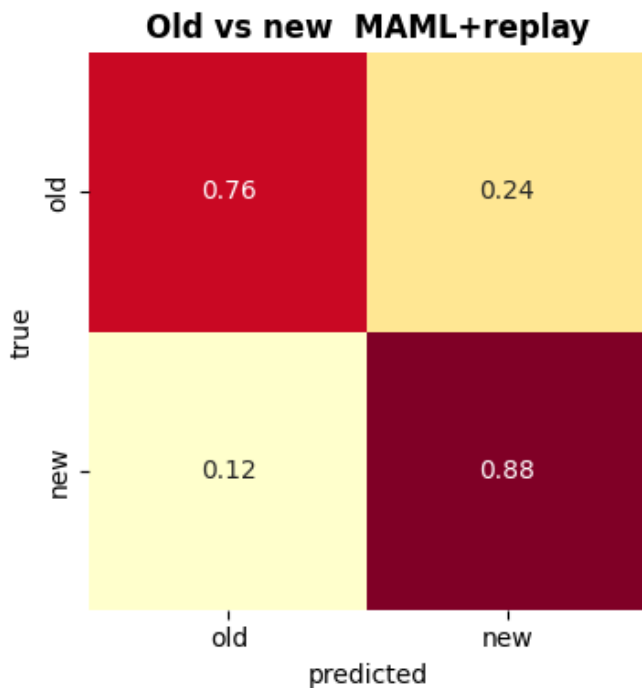


Fig. 7. Old-vs-new confusion for MAML+replay: most base clips are still predicted as base and most new clips as new. Without replay this matrix collapses (all mass predicted “new”).

opposite (base 0.677, new 0.295): re-exposure protects the old classes but the un-primed init learns the new sports slowly. *MAML+replay* combines both abilities—a fast-adapting init and re-exposure—so it is best balanced (0.60). The *fine-tune* control saw the same 30 sports, so *fine-tune+replay* is a strong baseline (0.53), but MAML+replay still beats it by 0.07, which is the clean evidence that *meta-learning itself*, not just the extra data, is responsible. The *LwF* variants over-protect: with only a few new clips, distilling the old outputs crushes the fresh new-class head, so their new accuracy is lower. Finally, the shot sweep (Fig. 5) shows the meta advantage is a property of the *init*: it is largest at 1 clip and shrinks as data grows—so MAML *complements* rehearsal rather than replacing it.

IX. THE ULAL CAPSTONE (TO BE RUN)

The final notebook, `ULAL.ipynb`, chains the best choice of every stage into one lifelong pipeline and is evaluated on real video. *This section describes the intended run; the numbers are expectations, not measurements.*

Stage by stage. (1) Train the ResNet50+LSTM teacher on the base classes (Stage 2 winner). (2) Grow it across the class stream with *replay*, the winning retention ability from Stage 3. (3) Distil the grown teacher into the MobileNet student (Stage 4), keeping most of the accuracy at $\sim 14\times$ fewer parameters for deployment. (4) Meta-train the compact student and then adapt it to brand-new classes from a handful of clips (Stage 5). (5) Test on real clips: for *known* classes this measures domain robustness (test the student directly); for *new* classes it measures learn-from-few (expand the head

and few-shot adapt, expecting accuracy to rise from near zero after only a few examples).

Why this order. Each stage supplies an ability the next needs—strong features, data-rich retention, deployable size, fast few-shot extension. The capstone is the claim that these abilities *compose* into one model that is accurate, small, retains old classes, and still picks up new ones from little data. We expect the student to keep most of its known-class accuracy on real clips and to improve markedly on new classes after few-shot adaptation, mirroring Stage 5. The measured numbers will be added once the run completes.

X. DISCUSSION AND LIMITATIONS

Three themes recur. *Measurement matters*: the group-disjoint split lowers but trusts every number, and in Stage 5 the harmonic mean reveals MAML+replay’s win that a base-dominated average hides. *Data beats cleverness for retention*: on video, rehearsal of a few clips outperforms every data-free regulariser (EWC, LwF). *Each tool has a regime*: distillation’s win is compression, and meta-learning’s win is few-shot extension—asked outside their regime (e.g. MAML with plenty of data) they tie simpler baselines.

Limitations: a single dataset and short streams (up to 20+5 classes); rehearsal stores clips, so its retention is not free (generative rehearsal is future work); meta-learning’s advantage is confined to the few-shot regime; the small student caps the achievable new-class accuracy; and the capstone’s real-video test depends on downloadable clips and is still pending.

XI. CONCLUSION

ULAL is a compact, honest, real-video pipeline for lifelong action recognition. A leakage-free split makes the numbers trustworthy; ResNet50+LSTM is a strong teacher (93.9%); rehearsal is the strongest continual learner (89.8%); distillation yields a $14\times$ smaller student at 82.7%; and, scored fairly, MAML+replay is the best-balanced few-shot adapter (0.60 harmonic), with a fine-tune control confirming the gain is from meta-learning. The capstone composes the best of each stage and is left to a final run.

AUTHOR CONTRIBUTIONS

Table VI indexes which author is responsible for which part.

TABLE VI
AUTHOR CONTRIBUTION INDEX.

Contributor	Sections
First A. Author	S1 Data split & leakage; S2 Backbone selection
Second B. Author	S3 Continual learning; S4 Knowledge distillation
Third C. Author	S5 Few-shot meta-adaptation; capstone; setup

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